

JADUAL 6
SEBUT HARGA
JADUAL SPESIFIKASI TERPERINCI

BIL	PERIHAL (SPESIFIKASI)		PATUH (√)		KETERANGAN TERPERINCI (WAJIB DIISI)
			YA	TIDAK	
A	PROJECT OVERVIEW				
	Project Title	Development of Advanced AI-Powered Badminton Performance Analysis System			
	Primary Objective	Transform current manual video analysis workflow from 3-4 hours per match to automated analysis completed within 30 minutes, while increasing coverage from 30-50% to 100% of all training sessions and competitions.			
	Target Users	1. National badminton coaches 2. Performance analysts 3. Elite national badminton players 4. Biomechanics and sports science staff			
B	FUNCTIONAL REQUIREMENTS: VIDEO/PERFORMANCE ANALYSIS				
1.	Player Tracking & Identification	Performance Requirements: - Accurately detect and track all 4 players (for doubles) and 2 players (for singles) simultaneously throughout a doubles/singles match without ID-swapping - Maintain tracking accuracy ≥95% under varying lighting conditions (indoor stadium lighting, outdoor natural light) - Generate real-time position coordinates for each player - Produce court coverage visualizations (heat maps) showing player movement patterns - Quantify court coverage efficiency as a percentage Input Requirements: - Must work with standard HD/4K video (360p minimum, 30-60fps) - Must handle both live streaming and recorded video files			

		• Must support common video formats (MP4, AVI, MOV) • Must maintain accuracy across various camera angles commonly used in badminton venues, including courtside, elevated, and corner positions • Should handle perspective distortion and varying camera heights without manual recalibration			
2.	Shuttle Detection & Trajectory Analysis	Performance Requirements: • Detect and track shuttle throughout rallies with $\geq 90\%$ accuracy • Measure shuttle velocity accurately for elite speeds (up to 500 km/h for smash shots) • Determine shuttle landing position on court with spatial accuracy of $\pm 20\text{cm}$ • Classify shot types with $\geq 85\%$ accuracy (minimum 12 types of shots: block, clear, defend, soft drive, hard drive, drop, lob, net, smash, tap, serve and receive serve) • Generate complete trajectory visualization for each shot			
3.	Tactical Pattern Recognition	Performance Requirements: • Automatically identify common tactical sequences (e.g., lift-smash-kill, drop-net, serve-return patterns) - Detect minimum 50 distinct tactical patterns commonly used in elite badminton • Classify rally phases (serve, build-up, finish) with $\geq 80\%$ accuracy • Distinguish between winning patterns and patterns leading to unforced errors and forced errors • Provide pattern frequency statistics and effectiveness metrics Doubles-Specific Requirements: • Recognize formation patterns (front-back, side-by-side, rotation in court) • Identify attacking vs defensive positioning • Detect strategy shifts during games Output Requirements: • Pattern library must be viewable and searchable by coaches • Coaches must be able to add custom pattern definitions			

4.	Rally & Match Statistics	Performance Requirements: The system must automatically extract and present: • Rally length (shots per rally, time duration) • Point outcomes (winner, unforced error, forced error) • Momentum score-time graph • Shot distribution by type and player • Service patterns and effectiveness - Return shot patterns • Third shot patterns • Scoring patterns (who scores in which situations) Statistical Accuracy: • Match statistics must have $\geq 95\%$ agreement with manual scoring • System must handle incomplete or partially obstructed rallies gracefully Real-time Capability: • Basic statistics must be available within 5 seconds of rally completion during live matches			
C	FUNCTIONAL REQUIREMENTS: BIOMECHANICS ANALYSIS				
1.	3D Human Pose Estimation	Performance Requirements: • Extract 3D skeletal positions from video without requiring markers, special suits, or specialized capture equipment • Minimum 17 keypoints tracked per person (full body including head, shoulders, elbows, wrists, hips, knees, ankles) • 3D joint position accuracy: Mean error $< 5\text{cm}$ compared to marker-based motion capture ground truth • Track skeletal positions at minimum 30fps, preferably 60fps • Handle occlusions and partial visibility gracefully • Support multi-person tracking (all 4 players in doubles and 2 players in singles) Input Flexibility: • Must work with standard match footage (no special camera setup required for basic analysis)			

2.	Joint Angle Analysis	Performance Requirements: ● Calculate key joint angles relevant to badminton technique: * Shoulder (internal/external rotation, abduction/adduction) * Elbow (flexion/extension) * Wrist (flexion/extension, ulnar/radial deviation) * Hip (flexion/extension, rotation) * Knee (flexion/extension, valgus/varus) * Ankle (dorsiflexion/plantarflexion) ● Provide angle measurements throughout the movement cycle (not just at static points) ● Identify peak angles and critical moments in technique execution Shot-Specific Analysis: ● System must analyze different shot types separately (smash technique differs from net shot technique) ● Provide shot-specific reference ranges based on elite player biomechanics Comparison Capability: ● Compare individual athlete technique to reference elite models - Track technique changes over time for individual athletes			
3.	Technique Visualization	Performance Requirements: ● Generate 3D skeletal animation overlays on original video ● Support side-by-side comparison of: * Athlete current technique vs their own previous technique * Athlete technique vs elite reference model * Two athletes performing same shot type ● Provide slow-motion playback with synchronized skeletal overlay ● Highlight specific joints or angles of interest ● Export visualization clips for coaching sessions Usability:			

		• Coaches must be able to select comparisons without technical expertise • Visualization must be clear and interpretable for athletes			
4.	Injury Risk Indicators	Performance Requirements: • Monitor biomechanical patterns associated with common badminton injuries: * Shoulder overuse/impingement (serving, overhead shots) * Knee injuries (landing mechanics, rotational stress) * Ankle sprains (landing position, stability) * Lower back strain (trunk rotation, asymmetry) Risk Indicators: • Joint loading asymmetry (left vs right side) • Landing mechanics quality scores • Repetitive movement patterns (overuse indicators) • Fatigue-related technique degradation Alert System: • Flag high-risk movements for coach review • Track injury risk scores over time • Generate reports for collaboration with physiotherapy staff IMPORTANT: • System is a decision support tool, NOT a medical diagnostic tool • All injury risk assessments require validation by qualified sports medicine professionals			

D	FUNCTIONAL REQUIREMENTS: OPPONENT INTELLIGENCE & TACTICAL SUPPORT				
1.	Opponent Profiling	Performance Requirements: ● Build comprehensive opponent profiles from historical match footage ● Profile must include: * Shot distribution patterns (where on court, which shots, when in rally) * Preferred tactical sequences * Weak zones (court areas with lower success rate) * Strength zones (court areas with higher success rate) * Service and return patterns * Performance under pressure (critical points, game points) * Formation preferences * Head-to-head record * Age * Handedness * Country ● Database Requirements: ● Support minimum 500 opponent profiles ● Each profile based on minimum 5 years of matches ● Automatic profile updates when new match data becomes available Data Sources: ● System must be capable of ingesting publicly available tournament footage ● Must handle varying video qualities from different sources			
2.	Performance Dashboards	Performance Requirements: <i>User-Specific Views:</i> ● Coaches & Analysts: Comprehensive team and individual athlete metrics, comparison tools, opponent analysis ● Athletes:			

		Personal performance tracking, technique feedback, training progress ● Analysts: Detailed statistical breakdowns, raw data export, custom query tools Dashboard Must Display: ● Match statistics (real-time and historical) ● Performance trends over time (daily, weekly, monthly, seasonal) ● Training session summaries ● Biomechanical metrics trends ● Comparison to personal best and team averages Performance: ● Dashboard page load time <2 seconds ● Real-time updates during live matches <5 seconds latency ● Support 100+ concurrent users without performance degradation Customization: ● Users can customize their dashboard layouts ● Ability to create custom metrics and visualizations Export: ● PDF reports (customizable templates) ● CSV data export for external analysis ● Video clip export with annotations			
3.	Tactical Recommendations	Performance Requirements: <i>Pre-Match Analysis:</i> ● Generate match-specific tactical recommendations based on: * Opponent profile analysis * Own team's strengths and weaknesses * Recent performance trends * Head-to-head historical data ● Identify top 3-5 tactical priorities			

		● Suggest specific shot patterns to exploit opponent weaknesses ● Recommend defensive strategies for opponent's strong patterns In-Match Support: ● Detect when opponent changes tactics mid-match ● Alert to emerging patterns or opportunities ● Track effectiveness of implemented tactics in real-time Post-Match Learning: ● Compare recommended tactics vs actual tactics used ● Analyze which recommendations were effective ● Improve future recommendations based on outcomes Presentation: ● Recommendations must be actionable and understandable by coaches ● Support visual presentation (videos and highlights)			
4.	Automated Match Preparation Reports	Performance Requirements: Report Generation: ● Automatically generate comprehensive pre-match scouting report when opponent is known ● Generation time: <10 minutes from request ● Report must include: * Opponent ranking and recent form summary * Key statistics/performance indicators comparison (our team vs opponent) * Weakness analysis with supporting video clips * Recommended tactical approach * Head-to-head analysis if previous matches exist Customization: ● Template must be customizable by coaching staff ● Coaches can add manual notes and annotations ● Support multiple language outputs (English, Malay, Mandarin)			

		Distribution: • PDF format with embedded video links • Email delivery • Mobile app access Video Integration: • Include short video clips demonstrating key points • Clips must be playable without requiring special software			
E	SYSTEM PERFORMANCE & QUALITY REQUIREMENTS				
1.	Processing Speed Requirements	Mandatory Performance Targets: <i>Batch Processing (Recorded Matches):</i> • Full match analysis (60-90 minutes of video): * Maximum 30 minutes processing time * Target: 20 minutes or less • Training session footage (30 minutes): * Maximum 15 minutes processing time <i>Real-Time Processing (Live Matches):</i> • Basic statistics and tracking: * Maximum 5 seconds latency from live action • Rally analysis: * Available within 30 seconds of rally completion • Basic biomechanics: * Available within 2 minutes of shot execution <i>Scalability:</i> • System must be capable of processing 100+ matches per week • Support concurrent processing of at least 5 matches simultaneously			

2.	Accuracy & Reliability Requirements	Minimum Acceptable Accuracy Levels: These are MANDATORY minimum thresholds. <i>Vendor solution must meet or exceed:</i> • Player Detection & Tracking: * ≥95% accuracy (no ID swaps during rally) • Shuttle Detection: * ≥90% accuracy • Shot Classification: * ≥85% accuracy across all shot types • 3D Pose Estimation: * Mean joint error <5cm vs marker-based ground truth • Tactical Pattern Recognition: * ≥80% agreement with expert human analysis • Match Statistics: * ≥95% agreement with manual scoring <i>System Reliability:</i> • Overall system uptime: * ≥95% during operational hours • Crash/failure rate: * <1% of processed videos • Graceful degradation: * System must handle corrupted or low-quality input without complete failure <i>Validation Requirements:</i> • Vendor must provide validation methodology and test results • ISN reserves right to conduct independent validation testing during UAT • Accuracy must be maintained across different lighting conditions, camera angles, and court types			
3.	Video Input Requirements	Format Compatibility: The system must accept: • Video formats: MP4, AVI, MOV, ,WebM (YouTube downloads), FLV (common broadcast formats)			

		● Codecs: H.264, H.265/HEVC, VP9 (YouTube), AV1 (newer videos) ● Frame rate: 24fps, 25fps, 30fps, 50fps, 60fps ● Maximum file size: Support videos up to 2 hours at 4K resolution ● Audio: system should work with audio track Resolution Support & Performance Tiers: ● Tier 1: 1080p and above * All accuracy targets specified in this document MUST be met * Full feature availability (tracking, biomechanics, tactical analysis) * This is the PRIMARY use case; system must be optimized for this ● Tier 2: 720p * Minimum 85% of stated accuracy targets * Core features operational (tracking, statistics, basic patterns) * Biomechanics may have reduced accuracy or limited keypoints ● Tier 3: 480p, 360p, 240p * System must accept without crashing or failing * System must warn user BEFORE processing: Example: "This video is 360p, expected performance player tracking 60-75%, shuttle detection unreliable, biomechanics unavailable. Analysis may take same time but produce limited results. Proceed?" * User decides whether degraded analysis is worthwhile * All outputs clearly labeled as "Low Resolution – Reduced Accuracy" ● Best-effort support for: * Player detection and basic positioning * Rally counting			
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		* Basic match statistics ● Not expected to work: * Shuttle trajectory and speed * Biomechanical analysis * Detailed tactical patterns <i>Compression & Quality Tolerance:</i> YouTube videos are heavily compressed. System must maintain performance with: ● High compression artifacts (blocking, banding) ● Variable bitrate encoding ● Watermarks and overlays (score graphics, broadcaster logos) ● Varying lighting conditions ● Different camera angles (broadcast cameras vs courtside) ● Moderate motion blur ● Partially obstructed views <i>Camera Angle Flexibility:</i> ● Maintain accuracy across various camera positions commonly found in badminton: * Courtside (low angle, parallel to net) * Elevated broadcast (high angle, 30-45 degrees) * Corner cameras (diagonal view) * End court cameras (behind baseline) ● Handle perspective distortion automatically ● Work with single-camera or multi-camera footage ● No manual recalibration required when switching between different tournament footage <i>Live Stream Input (Future Capability):</i> ● Support RTMP, HLS or RTSP streaming protocols ● Handle network latency and packet loss gracefully ● Buffer management for smooth processing <i>Upload Interface:</i>			
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		• Web-based upload with progress indicator • Drag-and-drop support • Batch upload capability (multiple files simultaneously) • Resume capability for interrupted uploads • Automatic resolution detection and tier classification before processing • YouTube URL import (optional but valued): Paste YouTube link, system downloads and process			
4.	User Interface Requirements	Usability Standards: • Maximum 3 clicks to reach any major feature from home screen • Onboarding tutorial for new users (integrated, skippable) • Context-sensitive help available throughout application • Error messages must be clear and actionable (not technical error codes) Accessibility: • Multilingual support: English, Bahasa Malaysia, Mandarin Chinese • Language switching without losing current work • Compliance with WCAG 2.1 Level AA accessibility standards • Keyboard navigation support • Responsive design (desktop, tablet, mobile) Platform Support: • Web application: * Latest 2 versions of Chrome, Firefox, Safari, Edge • Mobile app (optional but preferred): * iOS 15+, Android 10+ Performance: • Page load time: * <2 seconds on standard broadband connection • Video playback: * Smooth 30fps playback without stuttering • Interactive visualizations:			

		* <100ms response to user input			
F	SECURITY, PRIVACY & COMPLIANCE REQUIREMENTS				
1.	Access Control & Authentication	Mandatory Requirements: User Roles & Permissions: ● Role-Based Access Control (RBAC) with minimum roles: * System Administrator (full access) * Coaches (all analysis features, athlete data) * Performance Analyst (full analysis, reports, data export) * Athlete (own data only, limited features) * Viewer (read-only access to assigned content) AI Model Ownership & Usage ● Ownership of Results: * Any AI model "weights" or "fine-tuned parameters" (the intelligence the AI learns) created using ISN data belong exclusively to ISN. ● No Reuse: * The AI company is strictly prohibited from using these specific weights or settings for any other clients or projects. ● Data Isolation: * ISN's data and the resulting trained model must be kept separate from the company's other work to prevent "data leaking" into other products. ● Return of IP: * At the end of the project, the company must hand over all model files to ISN and delete their own copies. Authentication: ● Multi-factor authentication (MFA) required for staff accounts ● Single Sign-On (SSO) integration capability (SAML 2.0 or OAuth 2.0) ● Password complexity requirements enforced ● Account lockout after failed login attempts ● Session timeout after 30 minutes of inactivity Audit Logging:			

		• Log all user access, data modifications, exports • Logs must be tamper-proof and retained for minimum 1 year • Admin access to audit log viewer			
2.	Data Security	Encryption Requirements: • Data in Transit: TLS 1.3 or higher for all network communications • Data at Rest: AES-256 encryption for stored videos and sensitive data • Encryption key management: Secure key storage and rotation procedures Data Protection: • Regular automated backups (daily minimum) • Backup retention: Minimum 30 days • Disaster recovery plan with maximum 24-hour recovery time • Backup storage geographically separated from primary storage Security Testing: • Vendor must conduct security vulnerability assessment • Provide security testing report to ISN • Remediate any critical or high-severity vulnerabilities before deployment • Support ISN's own security audits if required			
3.	Privacy & Compliance	Regulatory Compliance: • Personal Data Protection Act 2010 (PDPA) Malaysia compliance • GDPR compliance (for international tournaments/athletes) • Data residency: Option to store all data within Malaysia if required Privacy Features: • Athlete consent management for data collection and use • Data anonymization capabilities for research/sharing • Right to erasure (delete athlete data upon request) • Data portability (export athlete data in standard format)			

		• Privacy policy and terms of service clearly presented to users Data Sharing Controls: • Granular control over what data is shared with whom • Temporary access links with expiration • Watermarking for exported videos/images • Download tracking and limitations			
G	INFRASTRUCTURE & INTEGRATION REQUIREMENTS				
1.	Deployment Options	Vendor Must Propose: the vendor must provide recommendations and cost comparison for: Option A • Cloud Deployment: * Fully managed cloud service (AWS, Azure, or GCP) • Automatic scaling and updates • Specify: Monthly operational costs, data egress costs, scaling limits Option B • On-Premises Deployment: * Self-hosted on ISN infrastructure • Complete hardware specifications and costs • Specify: Server requirements, network requirements, power/cooling Option C • Hybrid Deployment: * Combination of cloud and on premises • Specify: Which components on-premises vs cloud, data flow, costs • Vendor must clearly explain: * Pros and cons of each approach * Total Cost of Ownership (TCO) for 5 years * Performance implications * Data sovereignty considerations			

2.	System Integration	Mandatory Integration Requirements: ISN Infrastructure: • Must integrate with existing ISN network and IT infrastructure • Support ISN's IT security policies and procedures • Compatible with ISN's firewall and network configuration API Requirements: * RESTful API for third-party integrations * Comprehensive API documentation * API authentication and rate limiting * Webhook support for event notifications Data Export: • Standard data formats (CSV, JSON, XML) • Bulk export capability • Scheduled automated exports • Integration with common data analysis tools (Excel, Python, R) Future Integration Possibilities: • Vendor should design system to accommodate: * Wearable sensors integration (heart rate, GPS) * Court sensors (force plates, pressure sensors) * Other ISN systems (athlete management, medical records)			
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H	TRAINING, DOCUMENTATION & SUPPORT				
1.	Knowledge Transfer & Training	Training Requirements: <i>End User Training:</i> • Coaches training: Minimum 2 full-day sessions • Athletes training: Minimum 1 half-day session • Analysts training: Minimum 3 full-day sessions (technical depth) • Training materials in English and Bahasa Malaysia • Hands-on practice with real data • Video tutorials for self-paced learning Technical Staff Training (Transfer of Technology - TOT): • IT administrators: System administration, monitoring, troubleshooting • Biomechanics staff: Understanding AI outputs, validation procedures • Minimum 5 full days of technical training • Include: * System architecture overview * Maintenance procedures * Common issue troubleshooting * Data annotation and model retraining (if applicable) Certification: • Training completion certificates • Competency assessment Ongoing Support: • Quarterly refresher sessions during warranty period • On-demand training for new staff			
2.	Documentation Requirements	1. Mandatory Documentation (2 copies each): User Manuals: • Coach User Guide • Athlete User Guide • Analyst User Guide • Available in: English and Bahasa Malaysia			

		• Format: PDF + Online help system 2. Technical Documentation: • System Architecture Document • Database Schema and Design • API Documentation (interactive, e.g., Swagger/OpenAPI) • Deployment Guide • System Administration Manual • Troubleshooting Guide 3. AI/ML Documentation: • Model Architecture Descriptions • Training Methodology • Performance Validation Reports • Limitations and Known Issues • Model Update Procedures 4. Testing Documentation: • User Acceptance Test (UAT) Plan and Results • Security Test Results • Performance Test Results 5. Documentation Quality: • Clear, well-organized, searchable • Screenshots and diagrams where helpful • Version controlled and dated			
3.	Support & Maintenance	Warranty Period: 1 year from Go-Live Date Support Channels: • Email support: Response within 4 working hours • Phone support: Available during office hours (9am-5pm, Mon-Fri) • Ticketing system for issue tracking • Remote access support (with proper security protocols) - Bug Fix SLA (During Warranty): Critical Bugs (System down, data loss, AI processing pipeline failure): • Response: <4 hours			

		- Resolution: <24 hours High Priority (Major feature broken, significant accuracy degradation): - Response: <8 hours - Resolution: <3 working days Medium Priority (Minor feature issues, UI problems): - Response: <1 working day - Resolution: <5 working days Low Priority (Cosmetic issues, enhancement requests): - Response: <3 working days - Resolution: As agreed System Updates: - Security patches: Provided immediately when available - Bug fixes: Monthly update cycle - AI model improvements: Minimum twice yearly during warranty - Feature enhancements: As agreed based on user feedback All fixes and updates must be: - Documented in writing - Tested before deployment - Deployed during agreed maintenance windows			
I	PROJECT DELIVERY & ACCEPTANCE				
1.	Project Timeline	Total Duration: 24 Months (January 2026 - December 2027) Vendor must provide detailed project plan including: - Phase-by-phase breakdown with milestones - Resource allocation plan - Risk assessment and mitigation strategies - Dependencies and critical path - Deliverables schedule			

		Minimum Required Phases: Phase 1 (Months 1-3): Requirements & Foundation • Finalize detailed specifications • Infrastructure setup • Data collection and annotation protocols • Initial dataset preparation Phase 2 (Months 4-8): Core Development • AI model development and training • Core analysis features implementation • Basic user interface • Alpha testing Phase 3 (Months 9-12): • Advanced Features • Biomechanics module completion - Intelligence features • Integration testing • Beta testing with real users Phase 4 (Months 13-18): Refinement & Deployment • Performance optimization • User feedback incorporation • UAT with ISN staff • Production deployment • User training Phase 5 (Months 19-24): Stabilization & Expansion • System monitoring and optimization • Advanced features based on usage - Documentation finalization • Knowledge transfer completion			
2.	Acceptance Criteria	User Acceptance Testing (UAT): <i>The system will be accepted only when ALL the following are met:</i> 1. Functional Completeness:			

		• All features specified in this document are implemented and working • 100% of critical features pass UAT • Minimum 95% of high-priority features pass UAT 2. Performance Targets: • All accuracy targets met (verified through independent testing) • All processing speed requirements met • System stability: <1% failure rate during UAT period 3. Documentation: • All required documentation delivered and approved • User manuals validated by actual users • Technical documentation validated by IT staff 4. Training: • All training sessions completed • Minimum 80% of trained users rate training as 'Good' or better • ISN staff able to perform basic troubleshooting independently 5. UAT Process: • Minimum 30-day UAT period • ISN will provide test scenarios and acceptance criteria • Vendor must fix all critical and high-priority issues before acceptance • UAT success requires sign-off from: * Coaches * Performance Analysis Lead * IT Director * Project Sponsor			
3.	Payment Milestones (Indicative)	<i>Vendor should propose payment schedule aligned with deliverables. Suggested Structure (Vendor may propose alternative):</i>			

		• Contract Signing: 10% • Phase 1 Completion (Requirements & Foundation): 10% • Phase 2 Completion (Core Development): 20% • Phase 3 Completion (Advanced Features): 20% • Phase 4 Completion (Deployment & Training): 20% • UAT Acceptance & Go-Live: 15% • 6 Months Post-Implementation (Warranty mid-point): 5% Payment tied to deliverables: • Each payment requires ISN sign-off on completed milestone • Milestone acceptance based on predefined criteria • Retention percentage held until warranty completion Vendor must specify: • Detailed payment milestone breakdown • Deliverables associated with each payment • Acceptance criteria for each milestone			
J	VENDOR QUALIFICATIONS & PROPOSAL REQUIREMENTS				
1.	Vendor Experience	Mandatory Requirements: <i>Company Experience:</i> • Minimum 3 years' experience in AI/Computer Vision development • Demonstrated experience in sports analytics or motion analysis (provide case studies) • Minimum 2 completed projects of similar scope and complexity Reference Projects: Vendor must provide details for minimum 3 reference projects from past 5 years: • Project description and scope • Client name and contact information • Project duration and budget • Technologies used • Outcomes and client satisfaction • Lessons learned and how they apply to ISN project			

		Preferred Qualifications (Bonus Points): • Experience specifically with badminton or racquet sports • Published research in computer vision, sports analytics, or biomechanics • Existing partnerships with sports federations or national sports institutes • ISO certifications or quality management standards			
2.	Project Team Requirements	Minimum Team Composition: 1. Project Manager: • PMP or equivalent certification preferred • Minimum 5 years project management experience • Experience managing AI/ML projects 2. Machine Learning Engineers (Minimum 2): • Advanced degree in Computer Science, AI, or related field • Minimum 3 years' experience in computer vision • Proven track record in object detection, tracking, pose estimation 3. Software Engineers (Minimum 2): • Minimum 3 years full-stack development experience • Experience with video processing systems • Database design and optimization skills 4. Sports Scientist / Biomechanics Expert (Minimum 1): • Degree in Sports Science, Biomechanics, or Kinesiology • Understanding of badminton technique and performance analysis • Bridge between technical team and sports domain 5. UI/UX Designer (Minimum 1): • Portfolio demonstrating user-centered design • Experience with data visualization and analytics dashboards 6. For Each Key Team Member, Provide:			

		• Resume/CV • Role and responsibilities • Time commitment to project (%) • Relevant certifications • Previous similar projects			
3.	Technical Proposal Requirements	<i>Vendor's technical proposal must include:</i> 1. Solution Architecture: • High-level system architecture diagram • Technology stack justification (why these choices?) • Data flow diagrams • Integration approach with ISN infrastructure 2. AI/ML Approach: • Methodology for each AI module (player tracking, shuttle detection, pose estimation, etc.) • Training data requirements and acquisition plan • Validation methodology to demonstrate accuracy targets • Continuous improvement plan for AI models 3. Implementation Strategy: • Detailed project plan with Gantt chart • Resource allocation plan • Risk analysis and mitigation strategies • Quality assurance approach 4. Proof of Concept (Optional but Highly Valued): • Demo using sample badminton footage • Shows core capabilities (tracking, analysis, visualization) • Demonstrates vendor's understanding of badminton domain 5. Infrastructure Recommendation: • Cloud vs on-premises analysis with pros/cons • Hardware specifications if on premises			

		- Estimated operational costs (5-year projection) 6. Deployment & Training Plan: - Deployment strategy (big bang vs phased) - Training curriculum and materials - Knowledge transfer methodology 7. Post-Implementation Support: - Support model during and after warranty - Pricing for extended support contracts - Roadmap for future enhancements			
4.	Financial Proposal Requirements	Vendor must provide: 1. Detailed Cost Breakdown: - Development costs (by phase/module) - Infrastructure costs (hardware, cloud, licenses) - Training and documentation costs - Travel and accommodation (if applicable) - Contingency allowance 2. Total Cost of Ownership (5 Years): - Initial implementation (Year 0) - Annual operational costs (Years 1-5) - Support and maintenance costs - Infrastructure scaling costs 3. Payment Schedule: - Milestone-based payment plan - Percentage per milestone - Deliverables required for payment 4. Pricing for Optional Items: - Extended warranty beyond 1 year - Additional user licenses - Feature enhancements post-implementation - Mobile application (if not in base proposal)			

		• Integration with additional sensors/equipment 5. Commercial Terms: • Intellectual Property ownership • License model (perpetual, subscription, etc.) • Warranty terms • Penalty clauses for delays • Termination conditions			
K	EVALUATION CRITERIA				
	Proposal Evaluation	Technical Solution (40%): - Proposed approach to meeting functional requirements - Innovativeness and technical soundness - Demonstration of AI/ML capabilities - Scalability and futureproofing - Quality of proof-of-concept (if provided) Company & Team Qualifications (25%): - Relevant experience and track record - Quality of reference projects - Team qualifications and composition - Financial stability of vendor Project Approach (20%): - Project plan realism and detail - Risk management strategy - Knowledge transfer methodology - Support and maintenance plan Cost (15%): - Competitiveness of pricing - Value for money (not just lowest price) - Transparency of cost breakdown - Total cost of ownership ISN RESERVES THE RIGHT TO:			

		- Request clarifications or additional information - Conduct technical presentations - Negotiate terms with shortlisted vendors Award to vendor with best overall value (not necessarily lowest price)			
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